

Analysis of the Effect of On-the-Job Training on Employee Performance at the State Department for Roads

Abstract

This study investigates the impact of on-the-job training methods—coaching, job rotation, mentorship, and job shadowing—on employee performance at the State Department for Roads. Based on survey data from 159 employees, collected via Likert-scale questionnaires, the analysis employs reliability tests, descriptive statistics, correlation, and regression analysis using SPSS Version 30. Results indicate that all training methods positively influence performance, with coaching being the most effective. Qualitative insights complement the quantitative findings, offering practical recommendations for enhancing training programs.

1 Introduction

This study examines the impact of on-the-job training methods—coaching, job rotation, mentorship, and job shadowing—on employee performance at the State Department for Roads. Data were collected from 159 employees out of a sample of 173, using structured questionnaires with a Likert scale (1 = strongly disagree, 5 = strongly agree). The primary objective is to assess how these training methods affect performance and identify the most effective approach.

2 Methodology

2.1 Data Source

Data were gathered using structured questionnaires with Likert-scale responses.

2.2 Analysis Tool

SPSS Version 30 was used for all statistical analyses.

2.3 Methods

- Reliability analysis (Cronbach's Alpha)
- Descriptive statistics (means, standard deviations)
- Correlation analysis (Pearson's correlation)

- Multiple regression analysis

2.4 Variables

- **Independent:** Coaching, Job Rotation, Mentorship, Job Shadowing
- **Dependent:** Employee Performance

3 Results

3.1 Reliability Analysis

Cronbach's Alpha was used to assess the internal consistency of each construct. All values exceeded 0.7, indicating good reliability.

Table 1: Reliability Analysis

Construct	Cronbach's Alpha	Number of Items
Coaching	0.85	6
Job Rotation	0.78	6
Mentorship	0.82	6
Job Shadowing	0.80	6
Performance	0.88	6

3.2 Descriptive Statistics

Composite mean scores summarize the data for each construct.

Table 2: Descriptive Statistics for Composite Scores

Variable	Mean	Standard Deviation	N
Coaching	3.50	0.80	159
Job Rotation	3.20	0.70	159
Mentorship	3.40	0.90	159
Job Shadowing	3.30	0.80	159
Performance	3.60	0.70	159

3.3 Correlation Analysis

Pearson's correlation analysis revealed significant positive relationships between all training methods and performance.

3.4 Regression Analysis

Multiple regression analysis showed that the training methods explain 60% of the variance in performance ($R^2 = 0.60$).

Table 3: Demographic Characteristics

Variable	Category	Frequency	Percentage
Gender	Male	90	56.6%
	Female	69	43.4%
Age	18-27	20	12.6%
	28-37	50	31.4%
	38-47	40	25.2%
	48-57	30	18.9%
	Over 58	19	11.9%

Table 4: Correlation Matrix

	Coaching	Job Rotation	Mentorship	Job Shadowing	Performance
Coaching	1	0.45**	0.38**	0.42**	0.55**
Job Rotation	0.45**	1	0.30**	0.35**	0.48**
Mentorship	0.38**	0.30**	1	0.40**	0.52**
Job Shadowing	0.42**	0.35**	0.40**	1	0.50**
Performance	0.55**	0.48**	0.52**	0.50**	1

**p < 0.01

4 Discussion

The quantitative results confirm that all four training methods positively affect employee performance, with coaching showing the largest effect ($B = 0.25$). This aligns with prior research (?) emphasizing the value of tailored feedback. Job rotation, despite its significance, had the smallest effect ($B = 0.15$), possibly due to adaptation challenges (?). Qualitatively, coaching excels due to its focus on individualized development, while mentorship and job shadowing enhance knowledge transfer and practical exposure. Job rotation may require better planning to maximize its impact.

5 Conclusion

On-the-job training significantly enhances employee performance, with coaching being the most effective method, followed by mentorship, job shadowing, and job rotation. Recommendations include strengthening coaching programs with regular feedback, optimizing job rotation durations, and leveraging mentorship and job shadowing for skill development.

Note: The data presented are hypothetical. Actual SPSS output should replace the mock tables.

Table 5: Regression Analysis

Predictor	B	SE	t	p
Constant	1.20	0.30	4.00	<0.001
Coaching	0.25	0.05	5.00	<0.001
Job Rotation	0.15	0.04	3.75	<0.001
Mentorship	0.20	0.05	4.00	<0.001
Job Shadowing	0.18	0.04	4.50	<0.001